

Unit 1

****1 Mark Questions with Answers****

**Introduction to Python**

1. ****Q:**** Who developed Python?
****A:**** Guido van Rossum.
2. ****Q:**** In which year was Python released?
****A:**** 1991.
3. ****Q:**** Is Python compiled or interpreted?
****A:**** Interpreted.
4. ****Q:**** Write one feature of Python.
****A:**** Simple and easy-to-learn syntax.
5. ****Q:**** Mention one popular application area of Python.
****A:**** Web development.

**Applications of Python**

6. ****Q:**** Which library is used for Machine Learning in Python?
****A:**** Scikit-learn / TensorFlow.
7. ****Q:**** Name one Python framework for web development.
****A:**** Django or Flask.
8. ****Q:**** Which library is used for data analysis?
****A:**** Pandas.
9. ****Q:**** Name one GUI toolkit in Python.
****A:**** Tkinter.
10. ****Q:**** Which Python library is used for scientific computing?
****A:**** NumPy.

**Installation & IDEs**

11. ****Q:**** From which website can you download Python?
****A:**** www.python.org.
12. ****Q:**** What does IDE stand for?
****A:**** Integrated Development Environment.

13. **Q:** Write any two Python IDEs.
A: PyCharm, Jupyter Notebook.
14. **Q:** Which IDE comes by default with Python?
A: IDLE.
15. **Q:** Command to check Python version?
A: `python --version`.

Python Syntax, Indentation & Comments

16. **Q:** What is used to define code blocks in Python?
A: Indentation.
17. **Q:** Which symbol is used for single-line comments?
A: `#`.
18. **Q:** How do you write multi-line comments?
A: Using triple quotes `'''` or `"""`.
19. **Q:** Write a simple if statement in Python.
A:

```
python
if True:
    print("Hello")

```

20. **Q:** What error occurs if indentation is missing?
A: IndentationError.

Variables, Data Types & Type Casting

21. **Q:** How do you create a variable in Python?
A: By assigning value directly, e.g., `x = 10`.
22. **Q:** Can a variable hold values of different types in Python?
A: Yes.
23. **Q:** Data type of `3.14` is?
A: float.
24. **Q:** Data type of `"Hello"` is?
A: string.
25. **Q:** What is the output of `float(5)`?
A: 5.0.

****Operators****

26. ****Q:**** Which operator is used for exponentiation?

****A:**** `\*\*`.

27. ****Q:**** Difference between `==` and `is`?

****A:**** `==` checks values, `is` checks identity.

28. ****Q:**** Write one logical operator.

****A:**** `and`.

29. ****Q:**** Which operator checks membership?

****A:**** `in`.

30. ****Q:**** What does `&` do in Python?

****A:**** Bitwise AND.

****Input/Output Functions****

31. ****Q:**** Which function is used to take input from user?

****A:**** `input()`.

32. ****Q:**** Write a print statement.

****A:**** `print("Hello World")`.

33. ****Q:**** How to print multiple values in one line?

****A:**** Using commas, e.g., `print("A", "B")`.

34. ****Q:**** Which function converts input to integer?

****A:**** `int()`.

35. ****Q:**** What is the default separator in `print()`?

****A:**** Space.

****Control Structures****

36. ****Q:**** Write syntax of `if` statement.

****A:****

if condition:
statement

37. ****Q:**** Which statement is used to check multiple conditions?

****A:**** if-elif-else.

38. ****Q:**** Example of nested if?

****A.****

```
if x > 0:  
    if x % 2 == 0:  
        print("Positive Even")
```

39. ****Q.**** Which statement executes if condition is false?
****A.**** else.

40. ****Q.**** Why is elif used?
****A.**** To check multiple conditions when `if` fails.

**Looping**

41. ****Q.**** Which loop is used to iterate over a sequence?
****A.**** for loop.

42. ****Q.**** Print numbers first 10 natural numbers using for loop.
****A.****

```
for i in range(1, 11):  
    print(i)
```

43. ****Q.**** Which keyword is used to exit a loop?
****A.**** break.

44. ****Q.**** Which keyword skips current iteration?
****A.**** continue.

45. ****Q.**** What does `pass` do?
****A.**** Does nothing (placeholder).

46. ****Q.**** Can loops be nested in Python?
****A.**** Yes.

47. ****Q.**** Print even numbers from 2 to 10 using while loop.
****A.****

```
i = 2  
while i <= 10:  
    print(i)  
    i += 2
```

48. ****Q.**** What happens if condition in while loop never becomes false?
****A.**** Infinite loop.

49. ****Q.**** Which statement is used to terminate loop immediately?
****A.**** break.

50. **Q:** What happens if we use else without if?
A: `SyntaxError`.

2-Mark Questions with Answers

1. List any two important features of Python.

- Easy to learn and use (simple syntax).
- Interpreted and dynamically typed.

2. Give two examples of real-world applications of Python.

- Web development (Django, Flask).
- Data science and machine learning (NumPy, pandas, scikit-learn).

3. Differentiate between compiler and interpreter in Python installation.

- Compiler translates the whole program at once, while **Python uses an interpreter** that executes line by line.

4. What is the role of an IDE in Python programming?

- IDE provides an environment to **write, run, debug, and test Python code** easily (e.g., PyCharm, VS Code).

5. Write two rules for Python syntax and indentation.

- Indentation is mandatory for blocks.
- Statements do not end with semicolons by default.

6. Why are comments used in Python? Give an example.

- Comments make code readable and explain its purpose.
- Example: `# This is a single-line comment`

7. Differentiate between mutable and immutable variables in Python with examples.

- Mutable: Can be changed (e.g., `list = [1, 2, 3]`).
- Immutable: Cannot be changed (e.g., `tuple = (1, 2, 3)`).

8. Write examples of two numeric data types in Python.

- Integer → `x = 10` Float → `y = 3.14`

9. What is type casting? Give one example of implicit and explicit type casting.

- Type casting: Converting one data type to another.
- Implicit: `x = 5; y = 2.5; z = x + y` → result is float.

- Explicit: `int(3.14) → 3`

10. Write the difference between relational and logical operators in Python.

- Relational: Compare values (e.g., `>`, `<`, `==`).
- Logical: Combine conditions (e.g., `and`, `or`, `not`).

11. What is the use of the membership operator in Python? Give an example.

- Checks if a value exists in a sequence.
- Example: `"a" in "apple" → True`.

12. Write the difference between the identity operator `is` and equality operator `==`.

- `==` checks values `→ 10 == 10 → True`.
- `is` checks memory reference `→ [1, 2] is [1, 2] → False`.

13. Explain the role of bitwise operators with an example.

- Perform operations on bits.
- Example: `5 & 3 = 1` (binary AND).

14. What is the difference between `input()` and `print()` functions in Python?

- `input()` `→` reads user input.
- `print()` `→` displays output.

15. Write a program snippet using if-else to check if a number is even or odd.

```
n = 4
if n % 2 == 0:
    print("Even")
else:
    print("Odd")
```

16. What is the difference between if-elif-else and nested if statements?

- if-elif-else: Multiple conditions in sequence.
- Nested if: One `if` inside another.

17. Write a short program using a for loop to print the first 5 natural numbers.

```
for i in range(1, 6):
    print(i)
```

18. Write a while loop program to print numbers from 1 to 5.

```
i = 1
while i <= 5:
```

```
print(i)
i += 1
```

19. What is the use of the break statement in Python? Give an example.

- Used to exit a loop immediately.

```
for i in range(10):
    if i == 5:
        break
    print(i)
```

20. Differentiate between continue and pass statements with examples.

- **continue:** Skips current iteration.

```
for i in range(5):
    if i == 2:
        continue
    print(i)
```

- **pass:** Does nothing (placeholder).

```
for i in range(5):
    if i == 2:
        pass
    print(i)
```

5-Mark Questions with Answers

1. Explain the main features of Python with examples.

Answer: Elaborate the points

- **Simple & Easy** → Python has readable syntax.
- **Interpreted** → Executes line by line.
- **Portable** → Same code runs on Windows/Linux/Mac.
- **Dynamically Typed** → No need to declare data type.
- **Extensive Libraries** → Supports NumPy, Pandas, etc.

2. Write a short note on Python applications in real life.

Answer: Elaborate the points

Python is widely used in:

- **Web Development** – Django, Flask.
- **Data Science & AI** – TensorFlow, scikit-learn.
- **Game Development** – Pygame.
- **Automation/Scripting** – Writing automation scripts.
- **Networking & Cybersecurity** – Packet analysis and security testing.

3. Compare compiler and interpreter. Why is Python called an interpreted language?

Answer: Elaborate the points

- **Compiler** translates the whole program into machine code before execution.
- **Interpreter** translates line by line at runtime.
- **Python uses an interpreter** → It executes one line at a time, making debugging easier but execution slower compared to compiled languages like C++.

4. Explain Python variables and data types with examples.

Answer:

- **Variable:** A named storage for values.

```
x = 10 # integer
```

```
y = "Hello" # string
```

- **Data Types in Python:**

i. Numeric (int, float, complex)

ii. Sequence (list, tuple, string)

iii. Set & Dictionary

iv. Boolean

5. Explain type casting in Python with examples.

Answer: Elaborate it

- **Type Casting** → Converting from one data type into another.
- **Implicit Casting:** Python automatically converts types.

Eg:1 # Integer to float

```
x = 5
```

```
y = 2.5
```

```
z = x + y # result = 7.5 (float)
```

Eg 2: Integer to complex

```
x = 5
```

```
y = 6 + 7j
```

```
result = x+y # result = 11 + 7j (Complex)
```

- **Explicit Casting:** Using built-in functions.

Eg1: float to integer

```
num = int(3.7) # result = 3
```

Eg2: int to string

```
text = str(123) # result = "123"
```

6. Explain control structures in Python with examples.

Answer:

- **if statement Eg:**

```
x = 10
```

```
if x > 5:
```

```
    print("Greater")
```

- **if-else statement Eg:**

```
x = 3
```

```
if x % 2 == 0:
```

```
    print("Even")
```

```
else:
```

```
    print("Odd")
```

- **if-elif-else Eg:**

```
marks = 75
```

```
if marks >= 90:
```

```
    print("Grade A")
```

```
elif marks >= 60:
```

```
    print("Grade B")
```

```
else:
```

```
    print("Grade C")
```

7. Explain the use of loops in Python with examples of for and while loops.

Answer:

- **for loop:** Iterates over a sequence.

```
for i in range(1, 6):  
    print(i)
```

- **while loop:** Runs until condition is false.

```
i = 1  
while i <= 5:  
    print(i)  
    i += 1
```

8. Differentiate between break, continue, and pass statements with examples.

Answer:

- **break** → exits loop immediately.

```
for i in range(5):  
    if i == 3:  
        break  
    print(i, end=" ")
```

Output: 0 1 2

- **continue** → skips current iteration.

```
for i in range(5):  
    if i == 3:  
        continue  
    print(i, end=" ")
```

Output: 0 1 2 4

- **pass** → does nothing (placeholder).

```
for i in range(5):  
    if i == 2:  
        pass  
    print(i, end=" ")
```

Output: 0 1 2 3 4

9. Explain Python operators with examples.

Answer: Elaborate them with examples

- **Arithmetic:** + - * / % // **
- **Relational:** > < >= <= == !=
- **Logical:** and, or, not
- **Assignment:** =, +=, -=

- **Membership:** `in`, `not in`
- **Identity:** `is`, `is not`
- **Bitwise:** `&`, `|`, `^`, `~`, `<<`, `>>`

10. Write a program to print a multiplication table of a given number using loops.

```
1 num = int(input("Enter a number: "))
2 for i in range(1, 11):
3     print(f"{num} x {i} = {num*i}")
```

Enter a number: 3

```
3 x 1 = 3
3 x 2 = 6
3 x 3 = 9
3 x 4 = 12
3 x 5 = 15
3 x 6 = 18
3 x 7 = 21
3 x 8 = 24
3 x 9 = 27
3 x 10 = 30
```

11. What is the use of “else” clause for loops. Explain with examples.

A) The **else** clause with loops in Python is a special feature that runs **only when the loop finishes normally** (i.e., *not interrupted by a break statement*)

```
for i in range(5):
    if i == 3:
        break
    print(i, end=" ")
else:
    print("Loop finished successfully")
```

Output: 0 1 2

- **continue** → skips current iteration.

```
for i in range(5):
    if i == 3:
        continue
```

```
    print(i, end=" ")
else:
    print("Loop finished successfully")
```

Output: 0 1 2 4
Loop finished successfully

More 5-Mark Questions:

Operators

11. With examples, illustrate the use of assignment and augmented assignment operators.

A)

```
x = 5 # assignment
x += 3 # augmented assignment (x becomes 8)
x, y = 4, 5 # Multiple assignment
```

12. Explain membership operators (`in`, `not in`) with examples.

```
X = [3, 5, 7, 6, 2, 4]
print(3 in x) # Output: True
print(6 not in x) # Output: True
```

13. Differentiate between identity operators (`is`, `is not`) and relational operators with examples.

```
x = [1, 2, 3]
y = [1, 2, 3]

print(x == y) # True → contents are the same
print(x is y) # False → stored at different memory locations
print(x is not y) # True
```

14. Explain logical operators in Python with examples.

```
a, b = True, False
print("a and b:", a and b) # False
print("a or b:", a or b) # True
print("not a:", not a) # False
```

15. Explain Python's arithmetic operators with examples.

```
x = -10
y = 4
result = x + y * 2
print(result) # -2
```

16. Discuss relational operators in Python with suitable programs.

```
a, b = 10, 20
print(a > b)    # False
print(a < b)    # True
print(a == b)   # False
print(a != b)   # True
```

Control Structures (Decision Making)

17. Write a Python program using “if statement” to check whether a number is positive.

```
a = 10
if (a >= 0) :
    print("The no is positive")
else:
    print("The no is negative")
```

Output: The no is positive

18. Write a program using if-elif-else to calculate grade of a student based on marks.

```
1 marks = int(input("Enter your marks (0-100): "))
2 if marks >= 90:
3     grade = "A+"
4 elif marks >= 80:
5     grade = "A"
6 elif marks >= 70:
7     grade = "B"
8 elif marks >= 60:
9     grade = "C"
10 elif marks >= 50:
11     grade = "D"
12 elif marks >= 40:
13     grade = "E"
14 else:
15     grade = "F (Fail)"
16 print("Your grade is:", grade)
```

```
Enter your marks (0-100): 68
Your grade is: C
```

19. Write a program using nested conditions to check whether a year is a leap year or not.

```
1 year = int(input("Enter a year: "))
2
3 if year % 4 == 0:          # First check: divisible by 4
4     if year % 100 == 0:    # If divisible by 100, then check further
5         if year % 400 == 0: # If divisible by 400 -> Leap Year
6             print(year, "is a Leap Year")
7         else:
8             print(year, "is NOT a Leap Year")
9     else:
10        print(year, "is a Leap Year")
11 else:
12    print(year, "is NOT a Leap Year")
13
```

Enter a year: 2016
2016 is a Leap Year

Looping

20. Program to print the Fibonacci sequence using for and while loop.

```
1 n = int(input("Enter n: "))
2 a, b = 0, 1
3 print("Fibonacci Series:", end=" ")
4 for i in range(n):
5     print(a, end=" ")
6     a, b = b, a+b
```

Enter N: 8
Fibonacci Series: 0 1 1 2 3 5 8 13

```
1 n = int(input("Enter n: "))
2 a, b = 0, 1
3 count = 0
4 while count < n:
5     print(a, end=" ")
6     a, b = b, a + b
7     count += 1
```

Enter n: 6

21. Write a program to calculate the factorial of a number using a ****while loop****.

```
1 num = int(input("Enter a number: "))
2 fact = 1
3 i = 1
4 while i <= num:
5     fact *= i
6     i += 1
7 print("Factorial is:", fact)
```

Enter a number: 6
Factorial is: 720

22. Write a program to find GCD of the given numbers.

```
1 a = 64
2 b = 4
3 print(f"The gcd of {a} and {b} is", end=" ")
4 while b:
5     a, b = b, a % b
6 print(a)
```

The gcd of 64 and 4 is 4